

TABLE II
CONTINUED

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of events	138	138	138	138	138	138	138	138
Number of observations	847,314	847,314	847,314	847,314	846,729	847,314	847,314	847,314
Number of workers in the sample	660,771	2,248,711	287,484	2,277,624	781,003	469,226	1,830,393	2,349,485
Sample	Less than high school	High school or less	Teen	Women	Black or Hispanic	High probability	Medium probability	Low probability

Notes. The table reports effects of a minimum wage increase by demographic groups based on the event study analysis (see [equation \(1\)](#)) exploiting 138 state-level minimum wage changes between 1979 and 2016. The table reports five-year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment, and wages for individuals without a high school degree (column (1)), for individuals with high school degree or less schooling (column (2)), for teens (column (3)), for women (column (4)), for black or Hispanic workers (column (5)). Columns (6)–(8) report the results for groups of workers with differential probability of being exposed to the minimum wage changes. We use the [Card and Krueger \(1995\)](#) demographic predictors to estimate the probability of being exposed (see the text for details). Column 6 shows the results for the workers who have a high probability of being exposed to the minimum wage increase, column (7) for the middle-probability group, and column (8) for the low-probability group. All specifications include wage bin-by-state and wage bin-by-period fixed effects. Regressions are weighted by state-quarter aggregated population of the demographic groups. Robust standard errors in parentheses are clustered by state; significance levels are *0.10, **0.05, ***0.01.

The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pretreatment total employment. The third row, the percentage change in average wages in the affected bins, ($\% \Delta W$), is calculated using [equation \(2\)](#) in Section 2.2. The fourth row, percentage change in employment in the affected bins, is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{b}$). The fifth row, employment elasticity with respect to the minimum wage, is calculated as $\frac{\Delta a + \Delta b}{\% \Delta W}$, whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{b - 1}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

literature, though we note that it is not statistically significant given a standard error of 0.134.²⁴

In addition, we examine the effects on groups of workers with differential probability of being exposed to the minimum wage changes. To determine the likelihood of exposure, we construct a prediction model analogous to [Card and Krueger \(1995\)](#). We use observations from three years prior to the 138 events that also lie outside any of the five-year post-treatment windows and estimate a linear probability model of having a wage less than 125% of the statutory minimum wage on a rich set of demographic predictors.²⁵ We use the estimated model to obtain predicted probabilities of being exposed to minimum wage increases for all individuals in the sample regardless of their actual employment status. We then use the predicted probabilities to sort individuals into three groups: a “high-probability” group that contains individuals in the top 10% of the predicted probability distribution; a “low-probability” group that contains workers in the bottom 50% of the predicted probabilities; and a middle group containing the rest.

As expected, the high-probability group shows a considerably larger bite ($\Delta b = -9.4\%$) than the middle group ($\Delta b = -2.0\%$) and the low-probability group ($\Delta b = -0.4\%$). At the same time, the employment elasticities are very similar across the

24. We note that the teen estimates are unrelated to a focus on low-wage jobs, since the benefits of focusing on employment changes around the minimum wage is small for groups where most workers are low-wage ones. In [Online Appendix Table A.10](#) we show that our event study estimates are close to 0 for teens even if we use overall teen employment. At the same time, the classic two-way fixed effect specification with log minimum wage (TWFE-logMW) generates a sizable negative estimate for teens and for overall employment as well. In [Section IV](#) we discuss this discrepancy and argue that the difference between our approach and TWFE-logMW are driven by how the two empirical models are affected by employment shocks in the 1980s and early 1990s. [Online Appendix Table G.7](#) shows that in the post-1992 period, there is little divergence in teen minimum wage elasticities across standard specifications (including the TWFE-logMW); none of the specifications suggest noticeable losses to teen employment, and the elasticities are no more negative than -0.03 .

25. We use the same predictors as in [Card and Krueger \(1995\)](#): all three-way interactions of nonwhite, gender, and teen indicators; all three-way interactions of nonwhite, gender, and age 20–25 indicators; an indicator for having less than high school education; continuous highest grade completed variable; a third-order polynomial in labor market experience; Hispanic ethnicity indicator; interactions of the education and experience variables with gender. [Cengiz \(2018\)](#) shows the predictions using this Card and Krueger model compare favorably with those from more sophisticated machine learning-based methods.

probability groups. It is worth mentioning that the most precise estimate of the own-wage employment elasticity reported in this article appears in [Table II](#), column (6), where we look only locally around the minimum wage and also focus on the high-probability group: the confidence interval rejects any value smaller than -0.251 and larger than 0.663 . Such a confidence interval is quite narrow and rejects many estimates in the literature—highlighting the gains from combining the demographic profiling approach of Card and Krueger with the approach based on low-wage jobs advanced herein. The employment elasticities for the other groups are similar in magnitude, though less precise.²⁶

Overall, these findings provide little evidence of heterogeneity in the employment effect by skill level; the lack of a reduction in overall low-wage jobs does not appear to mask a shift in employment from low-skill to high-skill workers.

2. By Industrial Sectors. Much of the literature has focused on specific sectors like restaurants, where the minimum wage is particularly binding—therefore making it easier to detect a clear effect on the sectoral wage. In contrast, by focusing on changes at the bottom of the distribution, our approach can recover employment and wage responses even in industries where only a small fraction of workers are directly affected by the minimum wage increase. This allows us to provide a more comprehensive assessment of the effect of the policy across a range of industries.

In [Table III](#) we report estimates for various sectors in the economy. We assign workers to tradeable and nontradeable sectors following [Mian and Sufi \(2014\)](#).²⁷ The table shows that the bite of

26. In [Online Appendix A](#) we show that if we estimate the impact of the events on the aggregate wage and employment outcomes for each of the three probability groups, we can obtain a clear wage effect only for the high-probability group—capturing only around 36% of all minimum wage workers. This highlights the value of focusing at the bottom of the wage distribution which allows us to get an overall estimate for all low-wage workers.

27. [Mian and Sufi \(2014\)](#) define “tradeable” industries as having either the sum of imports and exports exceeding \$10,000 per worker or \$500 million total; their “nontradeable” sector consists of a subset of restaurant and retail industries; “construction” consists of construction, real estate, or land development-related industries. We use the list in [Mian and Sufi \(2014\)](#) of four-digit NAICS industries and census industry crosswalks to categorize all the industries in the CPS for 1992–2016. In our sample the shares of employment are 13%, 14%, 10%, for tradeable, nontradeable, and construction, respectively. See more details in [Online Appendix E](#). Since consistent industrial classifications limit our sample to

TABLE III
IMPACT OF MINIMUM WAGES ON EMPLOYMENT AND WAGES BY SECTORS (1992-2016)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Missing jobs below new MW (Δb)	-0.019*** (0.004)	-0.016* (0.008)	-0.066*** (0.007)	-0.003 (0.002)	-0.011*** (0.003)	-0.101*** (0.015)	-0.033*** (0.003)	-0.017** (0.008)
Excess jobs above new MW (Δa)	0.020*** (0.003)	0.011 (0.008)	0.072*** (0.011)	0.005 (0.006)	0.011*** (0.002)	0.101*** (0.015)	0.041*** (0.010)	0.011 (0.009)
% Δ affected wages	0.058*** (0.011)	0.058 (0.073)	0.056*** (0.014)	0.097 (0.086)	0.056*** (0.013)	0.049*** (0.012)	0.060*** (0.021)	0.073 (0.078)
% Δ affected employment	0.008 (0.031)	-0.111 (0.136)	0.022 (0.037)	0.051 (0.163)	0.009 (0.044)	-0.001 (0.026)	0.062 (0.080)	-0.101 (0.145)
Employment elasticity w.r.t. MW	0.007 (0.027)	-0.056 (0.069)	0.060 (0.103)	0.019 (0.059)	0.005 (0.026)	-0.002 (0.117)	0.086 (0.111)	-0.052 (0.074)
Emp. elasticity w.r.t. affected wage	0.140 (0.523)	-1.910 (3.922)	0.387 (0.597)	0.530 (1.311)	0.166 (0.763)	-0.011 (0.542)	1.040 (1.058)	-1.385 (2.956)
Jobs below new MW ($\bar{b}_{-1}$)	0.087	0.050	0.270	0.036	0.057	0.434	0.136	0.050
% Δ MW	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600